

# Role of aluminum-containing adjuvants in antigen internalization by dendritic cells in vitro

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## Abstract

An important step in the induction of an immune response to vaccines is the internalization of antigens by antigen presenting cells, such as dendritic cells (DCs). Many current vaccines are formulated with antigens adsorbed to an aluminum-containing adjuvant. Following injection of the vaccine the antigens may either elute or stay adsorbed to the adjuvant surface. Antigens, which elute from the adjuvant surface, are internalized by dendritic cells through macropinocytosis while those that remain adsorbed are internalized with the adjuvant particle by phagocytosis. The relative efficiency of these two routes of internalization was studied. Alpha casein (AC) labeled with a green fluorescent dye was selected as the model antigen. In order to model vaccine antigens that elute from aluminum-containing adjuvants following administration, dendritic cells were incubated with a solution of fluorochrome-labeled alpha casein. To model vaccine antigens that do not elute from aluminum-containing adjuvants following administration, dendritic cells were exposed to fluorochrome-labeled alpha casein adsorbed to aluminum hydroxide adjuvant (AH). Alpha casein has eight phosphate groups and adsorbs to aluminum hydroxide adjuvant through ligand exchange. Alpha casein does not elute from aluminum hydroxide adjuvant upon exposure to cell culture media. The uptake of antigen by dendritic cells was determined at 0.5, 1, 2 and 3 h by confocal microscopy and flow cytometry. Dendritic cells internalized both alpha casein in solution and alpha casein adsorbed to aluminum hydroxide adjuvant. However, the mean fluorescence intensity of dendritic cells incubated with adsorbed alpha casein was four times greater than dendritic cells incubated with alpha casein in solution. In addition, the internalization of alpha casein was enhanced when the mean aggregate diameter of the adjuvant in the cell culture media was reduced from 17  $\mu\text{m}$  to 3  $\mu\text{m}$ . It was concluded that antigen internalization by dendritic cells was enhanced when the antigen remained adsorbed to the aluminum-containing adjuvant following administration and the aggregate size of the adjuvant was smaller than dendritic cells which are approximately 10  $\mu\text{m}$  in diameter.

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## 1. Introduction

The ability of aluminum-containing adjuvants to increase the immunogenicity of antigens was discovered in 1926 when Glenny et al. [1] found that injection of guinea pigs with diphtheria toxoid precipitated with potassium alum provided

greater protection from the toxin than injection of toxoid alone. They also observed that when the precipitate was filtered, the filtrate contained none of the toxoid. This led to the belief that adsorption of the antigen to aluminum-containing adjuvants prior to administration is essential for enhancement of immunogenicity. Regulatory agencies require evidence that the antigen is adsorbed by aluminum-containing adjuvants. For example, the World Health Organization recommends that at least 80% of diphtheria and tetanus toxoids

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be adsorbed by aluminum-containing adjuvants in the vaccine [2].

Vaccines experience two different environments during their manufacture and use: the formulation prior to administration and interstitial fluid following intramuscular or subcutaneous administration [3]. Chang et al. [4] performed a study to determine how adsorption in the formulation affects antibody production in rabbits. The surface charge of aluminum-containing adjuvant was modified to produce vaccines in which 3, 35 or 85% of the 4.8 mg dose of lysozyme was adsorbed. The three vaccines induced much higher antibody titers than a 4.8 mg lysozyme solution but all three vaccines induced the same antibody titer. When the three vaccines were mixed with sheep lymph fluid to simulate interstitial fluid, the degree of lysozyme adsorption in each vaccine changed to 40% within 1 h. Because all of the vaccines had the same degree of lysozyme adsorption in interstitial fluid and induced the same degree of immunopotential, it was concluded that immunopotential by aluminum-containing adjuvants correlates with the degree of antigen adsorption in interstitial fluid following administration rather than the degree of antigen adsorption in the vaccine.

A recent study [5] investigated whether it was necessary for the antigen to elute or remain adsorbed to aluminum hydroxide adjuvant (AH) in order to produce immunopotential. The model antigens used were alpha casein (AC), which completely adsorbed to AH through the ligand exchange mechanism, and dephosphorylated alpha casein (DPAC), which completely adsorbed to AH by electrostatic attraction [6]. After exposure to interstitial fluid for 1 h, 100% of the AC remained adsorbed to AH while only 10% of the DPAC remained adsorbed. All of the AC remained adsorbed to AH even after 24 h of exposure to interstitial fluid. In comparison, the DPAC was completely eluted after 6 h of exposure to interstitial fluid. It was concluded that the strong ligand exchange adsorption mechanism, whereby a phosphate in AC displaced a surface hydroxyl to form an inner sphere complex with AH [7], maintained complete adsorption of AC in interstitial fluid [8]. However, since DPAC adsorbed to AH by forming an outer sphere complex by electrostatic attraction [7], it completely eluted from AH during exposure to interstitial fluid. It was found that AH potentiated the immune response in mice for both AC and DPAC. The results indicated that aluminum-containing adjuvants potentiate the immune response when the antigen elutes from the adjuvant following administration as well as when the antigen remains adsorbed following administration.

An important step in the induction of the immune response is the internalization of the antigen by antigen presenting cells, such as dendritic cells (DCs). Macropinocytosis and phagocytosis are two mechanisms of antigen internalization by DCs [9–14]. Antigens that elute from the adjuvant surface are internalized by DCs by macropinocytosis while those that remain adsorbed are internalized by phagocytosis.

While the study with AC and DPAC [5] demonstrated that antigen internalization by antigen presenting cells occurs by

both macropinocytosis and phagocytosis, the relative efficiency of each route was not studied. Thus, this study was undertaken to compare antigen internalization by macropinocytosis and phagocytosis. AC labeled with a green fluorescent dye was chosen as the model antigen. AC contains eight phosphate groups, is adsorbed to AH by ligand exchange and does not elute from AH when exposed to interstitial fluid [5]. Dendritic cells were incubated with a solution of fluorescent-labeled AC to model antigens that elute from the adjuvant surface following administration. Dendritic cells were also incubated with fluorescent labeled AC adsorbed to AH to model antigens that do not elute following administration. In addition, the effect of the aggregate size of the adjuvant on the internalization of AC by phagocytosis was studied. The efficiency of antigen internalization was monitored by confocal microscopy and flow cytometry.

## 2. Materials and methods

### 2.1. Adjuvants

Aluminum hydroxide adjuvant (Rehydralgel HPA, Reheis, Berkeley Heights, NJ) and aluminum phosphate adjuvant (AP) (Adju-Phos, Brenntag Biosector, Elsenbakken, Denmark) were obtained commercially and diluted to 1 mg Al/ml with doubly distilled water (ddH<sub>2</sub>O). AH was modified by pretreatment with phosphate anion at P/Al molar ratios of 1:1 or 2:1. For the 1:1 phosphate-treated AH (PTAH-1), 769  $\mu$ l of commercial AH containing 13 mg Al/ml was diluted to 10 ml with 3.7 ml of 0.1 M Na<sub>3</sub>PO<sub>4</sub> adjusted to pH 7.4 and 5.531 ml of doubly distilled water adjusted to pH 7.4 and mixed with a magnetic stirrer for 1 h. The 2:1 phosphate-treated AH (PTAH-2) was prepared similarly except 7.4 ml of 1 M Na<sub>3</sub>PO<sub>4</sub> was used. PTAH-1 and PTAH-2 contained 1 mg Al/ml.

The zeta potential of each adjuvant was determined by Doppler electrophoretic light scattering analysis (DELSA 440, Coulter, Hialeah, FL). Each adjuvant was diluted 1:80 with the supernatant of the adjuvant obtained by centrifugation.

### 2.2. Adsorption isotherms

Each adjuvant prepared in Section 2.1 was diluted to 0.1 mg Al/ml with 10  $\mu$ M veronal buffer (Sigma, St. Louis, MO) and adjusted to pH 7.4 with 0.1 M HCl. AC solutions ranging from 100–300  $\mu$ g/ml were prepared and adjusted to pH 7.4 with 0.1 M NaOH or 0.1 M HCl. The adsorption isotherms were prepared by combining 100  $\mu$ l of the diluted adjuvant and 100  $\mu$ l of the appropriate AC solution and mixing using a vortex mixer at room temperature for 1 h. A preliminary experiment indicated that equilibrium was reached in less than 30 min. The mixtures were then centrifuged and the AC concentration of the clear supernatant determined by the BCA assay (Pierce, Rockford, IL). The adsorptive capac-

ity was calculated from the linearized form of the Langmuir equation as recently described [15]. The reported adsorptive capacities were the mean of three separate adsorption isotherms.

The adsorption isotherms of BODIPY FL labeled AC by AH and AP was determined as described above.

### 2.3. Fluorescent labeling of alpha casein

Alpha casein (Sigma, St. Louis, MO) was labeled with BODIPY FL (Molecular Probes, Eugene, OR). A 5 mg/ml solution of AC was prepared by dissolving 75 mg of AC in 14 ml of doubly distilled water (ddH<sub>2</sub>O), adjusting the pH to 8.3 with 0.1 M NaOH and adjusting the volume to 15 ml with ddH<sub>2</sub>O. The pH of the AC solution was adjusted to pH 8.3 because the reaction is most efficient between pH 7.5 and 8.3 [16] and addition of BODIPY FL will slightly lower the pH. For labeling, 1 ml of the AC solution was combined with 68 µl of 5 mg/ml BODIPY FL and mixed using a vortex mixer for 1 h at room temperature. The labeled AC was separated from unreacted dye by size exclusion chromatography using Sephadex G-50 fine (Sigma, St. Louis, MO) equilibrated with ddH<sub>2</sub>O. The BODIPY FL labeled AC eluted from the column first and was collected. The concentration of the labeled AC was determined by the BCA assay (Pierce, Rockford, IL) and was approximately 1 mg BODIPY FL labeled AC/ml.

### 2.4. Dendritic cell culture

Dendritic cells were derived from the bone marrow of 10–12 weeks old female BALB/c mice. The mice were anesthetized with 90 mg ketamine and 10 mg xylazine/kg and then euthanized. The femurs and tibias were collected in ice-cold Hank's balanced salt solution (HBSS, Sigma, St. Louis, MO) with 50 µg/ml gentamycin. The muscles were removed with a scalpel and by rubbing the bones with tissue. The ends of the bones were then cut off with scissors and crushed. The bone marrow was flushed out with 2–3 ml of media in a syringe with a 25-gauge needle. All bone marrow cells were collected and washed twice with HBSS. The bone marrow cells were cultured in 10 cm petri dishes containing approximately  $5 \times 10^5$  cells/ml in 10 ml total volume. The cells were maintained at 37 °C with 5% CO<sub>2</sub> in RPMI 1640 with 5% fetal calf serum (Sigma, St. Louis, MO), 0.05 µM β-mercaptoethanol (Sigma, St. Louis, MO), 3% supernatant of J558L cells transfected with granulocyte macrophage colony stimulating factor (GM-CSF), 2 mM L-glutamine, 100 U/ml penicillin, 100 µg/ml streptomycin and 0.25 µg/ml amphotericin B. After 3 days the nonadherent cells were collected and discarded and 10 ml of fresh media was added. On day 5, 5 ml of media was replaced with fresh media. On day 7, an additional 5 ml of media was added to the cells. On day 8, the cells were collected, washed and resuspended in the media without GM-CSF. The culture contained approximately 75% immature DCs. The study was approved by the Purdue University Animal Care and Use Committee.

### 2.5. Model vaccines

A 0.1 mg/ml solution of BODIPY FL labeled AC was prepared by diluting an appropriate volume of BODIPY FL labeled AC as prepared in Section 2.3 to 10 ml with ddH<sub>2</sub>O and adjusting the pH to 7.4. A 1 mg Al/ml suspension of each adjuvant was prepared as described in Section 2.1. The BODIPY FL labeled AC adsorbed to each adjuvant was prepared by combining 2.5 ml of 0.1 mg BODIPY FL labeled AC/ml with 2.5 ml of a 1 mg Al/ml suspension of each adjuvant and mixing with a magnetic stirrer for 1 h.

### 2.6. Treatment of DCs with AC

Immature DCs ( $1 \times 10^6$  cells/ml) were treated with a media alone, solution of BODIPY FL labeled AC (0.01 mg/ml) or a suspension of BODIPY FL labeled AC (0.01 mg/ml) adsorbed to each adjuvant (0.1 mg Al/ml). Treatment groups were set up in 6 well plates with a total volume of 3 ml/well. For the cells treated with media alone, 0.6 ml of media was added to 2.4 ml of DCs. For the DCs treated with a solution of AC, 0.3 ml of the 0.1 mg/ml BODIPY FL AC solution and 0.3 ml of media were added to 2.4 ml of DCs. For DCs treated with AC adsorbed to each adjuvant, 0.6 ml of each model vaccine was added to 2.4 ml of DCs. Following incubation for 0.5, 1, 2, or 3 h, the cells were fixed with 2% paraformaldehyde prior to analysis by confocal microscopy and flow cytometry.

### 2.7. Confocal microscopy

For analysis by confocal microscopy, 1 ml of treated DCs was washed twice in flow buffer (1 g BSA, 1 g sodium azide in 1 l phosphate buffered saline). After washing, 0.3 ml of 2% paraformaldehyde was added to each sample. The cells were stained with AlexaFluor 568 conjugated phalloidin (Molecular Probes, Eugene, OR). The phalloidin solution (0.1 ml) was diluted with 0.19 ml of PBS. The DCs were centrifuged at 1500 rpm ( $537 \times g$ ) for 4 min at 4 °C and the supernatant was removed. The DCs were then resuspended in 0.2 ml of diluted phalloidin and incubated at 4 °C for 20 min. Following incubation the DCs were washed twice with 0.25 ml PBS. Each sample was then observed by confocal microscopy at 60× magnification. The z-series images obtained were created with Confocal Assistant (Universal Imaging Corp., Downingtown, PA) and Adobe Photoshop 7.0® (Adobe, San Jose, CA).

### 2.8. Flow cytometry

For analysis by flow cytometry DCs were stained with an antibody to MHC class II molecules (PE labeled anti-mouse I-A/I-E {M5/114.15.2} Beckton Dickinson, Franklin Lakes, NJ). Two milliliters of DCs from Section 2.5 were centrifuged, resuspended in 0.2 ml of 10% normal rabbit serum with 1% bovine serum albumin in PBS and incubated for

15 min. The DCs were washed twice in flow buffer (from Section 2.9), resuspended in 0.2 ml of the antibody diluted 1:200 in flow buffer and incubated for 30 min in the dark. The cells were washed twice in flow buffer and fixed in 0.6 ml of 2% paraformaldehyde. Each sample was assayed for 4 min by flow cytometry (Epics XL, Coulter, Fullerton, CA) and the data were analyzed by WinMDI® (Joe Trotter, Scripps Institute, La Jolla, CA).

The data were analyzed by first comparing the forward scatter versus side scatter density plots for adjuvant only and for media-treated DCs. These plots give an indication of where the adjuvant population and DC population will be located in the samples that contain both adjuvant particles and DCs. In general, the DCs have a higher intensity forward scatter than adjuvants. A density plot of forward scatter versus side scatter of a sample of DCs treated with BODIPY FL labeled AC adsorbed to adjuvant was then constructed and a gate was made around the DC population. Next a plot of red fluorescence versus green fluorescence for the gated DC population was constructed. There were two populations observed; one at high red fluorescence and one at low red fluorescence. The high red fluorescence population were activated DCs which express high levels of MHC II. The low red fluorescence population were immature DCs which express low levels of MHC II [9]. Immature DCs are the cell population of interest because they are highly efficient at antigen capture. Therefore, a gate was made around that population and the geometric mean intensity of the green fluorescence was calculated for the gated population. This procedure was repeated for each time point of each sample.

## 2.9. Degree of adsorption of AC in cell culture media

The degree of adsorption of AC by each adjuvant following dilution with cell culture media was studied by adding 0.6 ml of each model vaccine (from Section 2.5) to 2.4 ml of cell culture media in a six well plate and mixing by gentle agitation to uniformly distribute the adjuvant. Samples were incubated at 37 °C without mixing. After 3 h, supernatants (100 µl) were obtained by centrifuging at 10,000 rpm (16,750 × *g*) for 15 min. The concentration of AC in the supernatants was determined by ELISA. Sheep anti-bovine casein (Bioscience International, Saco, ME) was used as the coating antibody, rabbit anti-bovine casein (Immunology Consultants Laboratory, Newburg, OR) was used as the capture antibody and goat anti-rabbit IgG conjugated to

horseradish peroxidase (Sigma, St. Louis, MO) was used as the detection antibody.

## 2.10. Aggregate size

The aggregate size of each adjuvant in cell culture media was determined by adding 0.6 ml of each model vaccine (from Section 2.5) to 2.4 ml of cell culture media in a six well plate and mixing by gentle agitation to uniformly distribute the adjuvant. Samples were incubated at 37 °C for 3 h and examined by fluorescent microscopy (p 1000, Nikon, Tokyo). The mean diameter of the adjuvant aggregates was determined by image analysis using Image-Pro Plus Version 5.0 (MediaCybernetics, Silver Spring, MD).

# 3. Results and discussion

## 3.1. Surface charge of adjuvants

Evidence for the adsorption of phosphate by AH in PTAH-1 and PTAH-2 is seen in Table 1 as the zeta potential of AH, which is normally positive [17], became increasingly negative as the concentration of phosphate used to pretreat AH increased. The negative surface charge of PTAH-2 approached that of AP [17].

## 3.2. Adsorption isotherms

Adsorption isotherms of AC by the four adjuvants were determined because AC has an iep of 4.6 [18] and is negatively charged at pH 7.4. Therefore, an electrostatic attractive force will exist between AC and AH while an electrostatic repulsive force will operate between AC and PTAH-1, PTAH-2 and AP. As seen in Table 1, AC was adsorbed by each adjuvant although the adsorptive capacity decreased as the concentration of phosphate used in the pretreatment increased. The adsorption of negatively charged AC by negatively charged PTAH-1, PTAH-2 and AP occurred in spite of electrostatic repulsive forces. Iyer et al. [19] recently reported that AC is adsorbed by AH and phosphate-treated AH by ligand exchange of phosphate groups of AC with surface hydroxyls of the adjuvant. AC adsorption by the four adjuvants in this study was probably also by ligand exchange. Because the concentration of AC in the model vaccines (0.1 mg AC/mg Al) was below the adsorptive capacity of each

Table 1  
Properties of aluminum-containing adjuvants

| Adjuvant                                | Zeta potential (mV) | Alpha casein adsorptive capacity (mg/mg Al) | Percentage alpha casein adsorption in cell culture media | Mean diameter of adjuvant aggregates in cell culture media (µm) |
|-----------------------------------------|---------------------|---------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|
| Aluminum hydroxide                      | +33.5               | 2.1                                         | 98                                                       | 17                                                              |
| Phosphate-treated aluminum hydroxide -1 | −14.4               | 1.5                                         | 99                                                       | 14                                                              |
| Phosphate-treated aluminum hydroxide -2 | −26.3               | 1.4                                         | 99                                                       | 10                                                              |
| Aluminum phosphate                      | −29.4               | 0.4                                         | 98                                                       | 3                                                               |

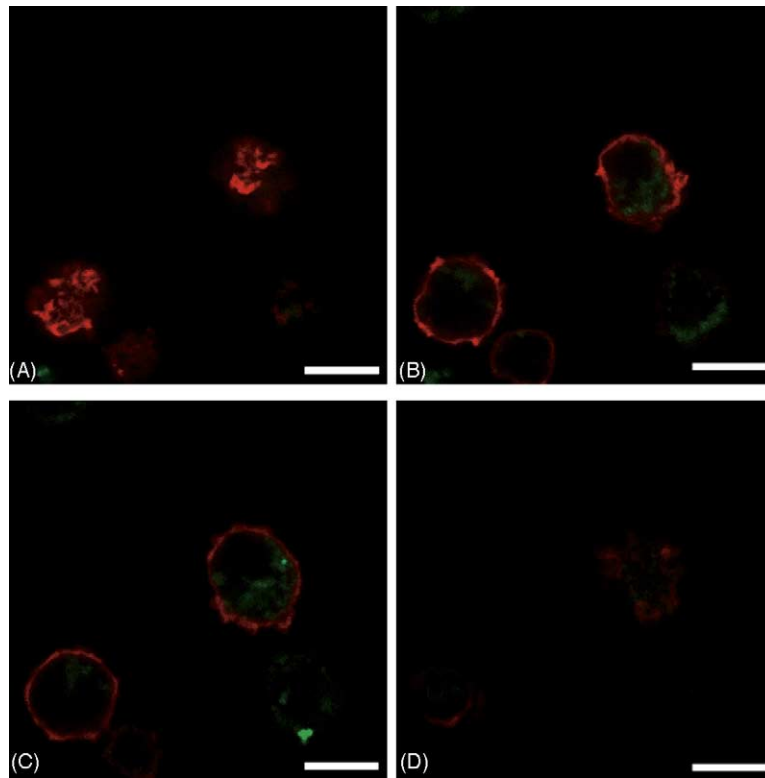


Fig. 1. Dendritic cells treated with a solution of BODIPY FL (green) labeled alpha casein. The cytoskeleton was stained red. The images are from a z-series of 43 slices. Key: (A) slice 4; (B) slice 13; (C) slide 26; (D) slice 37. The bars represent 10  $\mu$ m.

adjuvant, the AC was completely adsorbed in each model vaccine.

The mean adsorptive capacity of BODIPY FL labeled AC by AH (2.1 mg/mg Al) and AP (0.38 mg/mg Al) was essentially the same as the unlabeled AC, indicating that the fluorescent label did not affect adsorption of AC by the aluminum-containing adjuvants. Thus, the results of studies with BODIPY FL labeled AC can be applied to AC.

### 3.3. Elution of AC by cell culture media

The adsorbed AC did not elute when each model vaccine was exposed to cell culture media for 3 h (Table 1). Thus, any AC internalized by DCs from the model AC/adjuvant vaccines must occur by phagocytosis of the adjuvant particles.

### 3.4. Confocal microscopy

The cytoskeleton of DCs was stained with phalloidin to enable visualization of the red DCs and green AC. Visual evidence for the internalization of AC by macropinocytosis was obtained when DCs that had been exposed to a solution (10  $\mu$ g AC/ml) of BODIPY FL labeled AC were examined by confocal microscopy. Fig. 1A shows a focal plane at the top of two DCs. Only the red cytoskeleton of the DCs was visible. Fig. 1B and C show focal planes that are through the center of the DCs. In these images green fluorescence was

present within the red cytoskeleton of the DCs. The green materials are pinocytic vesicles that contain BODIPY FL labeled AC. Finally, Fig. 1D shows a focal plane at the bottom of the DCs and only the red cytoskeleton was observed. The z-series showed that AC in solution was internalized by the DCs. Thus, antigen that elutes from aluminum-containing adjuvants following intramuscular or subcutaneous administration can be internalized by macropinocytosis.

AH has a complex morphology [20,21]. The basic unit is fibrous primary particles having dimensions of 4.5 nm  $\times$  2.2 nm  $\times$  10 nm. However, the primary particles do not exist as solitary particles but form aggregates to minimize the surface free energy. AP also has a complex morphology [20]. The primary particles are platy having a diameter of approximately 50 nm. The primary particles aggregate to reduce the surface free energy. The aggregate diameters of the four adjuvant studied following adsorption of BODIPY FL labeled AC and exposure to cell culture media are presented in Table 1. The mean aggregate size decreased in the order AH > PTAH-1 > PTAH-2 > AP.

Alpha casein adsorbed to AH was also internalized by DCs. Fig. 2A shows a focal plane at the top of two DCs that were exposed to a suspension of BODIPY FL labeled AC adsorbed to AH. The red cytoskeleton of the top of the DCs was apparent. The green fluorescence of BODIPY FL labeled AC adsorbed to AH was also seen in the field. Fig. 2B and C show focal planes within the DCs. Green fluorescent material was seen within the red cytoskeleton of the DCs. The



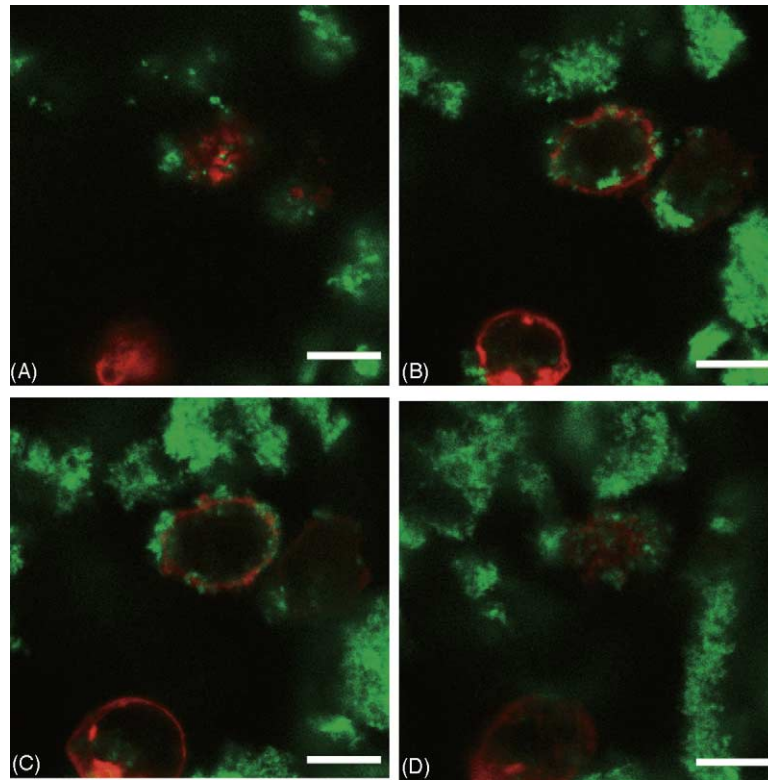


Fig. 2. Dendritic cells treated with BODIPY FL (green) labeled alpha casein adsorbed to aluminum hydroxide adjuvant. The cytoskeleton was stained red. The images are from a z-series of 46 slices. Key: (A) slice 3; (B) slice 15; (C) slice 27; (D) slice 40. The bars represent 10  $\mu\text{m}$ .

diameter of the DCs was approximately 10  $\mu\text{m}$  and the mean diameter of the BODIPY FL labeled AC/AH aggregates was 17  $\mu\text{m}$  (Table 1). Fig. 2B and C indicate that even though the diameter of the AH aggregates was larger than the DCs, the DCs were able to internalize portions of the AH aggregates. A recent study [22] showed that the primary particles in AH aggregates were loosely associated and deaggregation readily occurred. Fig. 2D shows a focal plane at the bottom of the DCs.

The images in Fig. 2 indicate that AC adsorbed to AH was internalized by DCs through phagocytosis. However, the DCs did not internalize entire AH aggregates. Rather, the DCs engulfed and internalized fragments of the AH aggregates.

### 3.5. Flow cytometry

The DCs were also analyzed by flow cytometry to quantify the internalization efficiency of AC in solution and AC adsorbed to AH. When the fluorescence intensities were measured, it was found that internalization of both AC in solution and AC adsorbed to AH occurred quickly as most internalization occurred during the first 30 min of exposure (Fig. 3). The DCs that were incubated with AC adsorbed to AH had a fluorescence intensity four times greater than DCs treated with the same amount of AC as a solution. This indicates that antigen internalization through phagocytosis of adsorbed antigen was more efficient than macropinocytosis of antigen in solution.

### 3.6. Effect of adjuvant aggregate diameter

The observation that enhanced internalization of adsorbed antigen occurred even though the adjuvant aggregates were larger than the DCs led to an experiment to study the effect of adjuvant aggregate size on the internalization of adsorbed

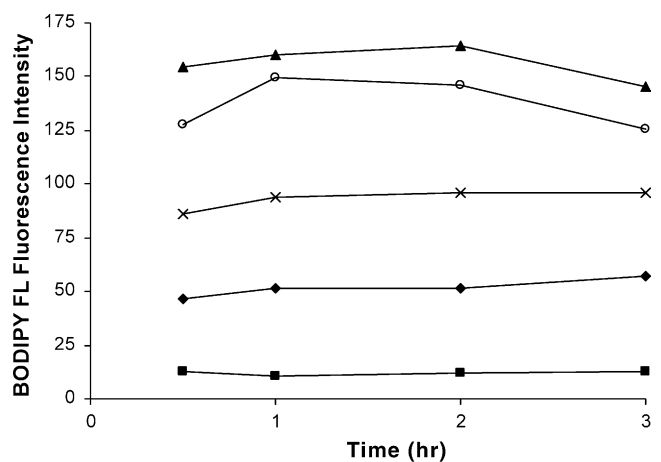


Fig. 3. BODIPY FL fluorescence intensity of dendritic cells treated with BODIPY FL labeled alpha casein in solution (■); BODIPY FL labeled alpha casein adsorbed to aluminum hydroxide adjuvant (◆); BODIPY FL labeled alpha casein adsorbed to phosphate-treated aluminum hydroxide adjuvant-1 (×); BODIPY FL labeled alpha casein adsorbed to phosphate-treated aluminum hydroxide adjuvant-2 (○); BODIPY FL labeled alpha casein adsorbed to aluminum phosphate adjuvant (▲).

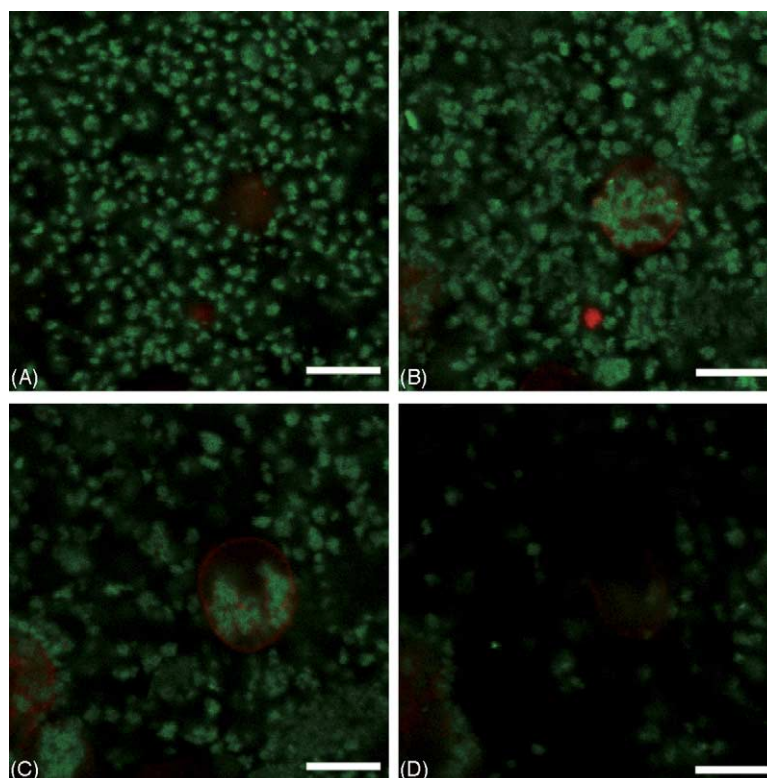


Fig. 4. Dendritic cells treated with BODIPY FL (green) labeled alpha casein adsorbed to aluminum phosphate adjuvant. The cytoskeleton was stained red. The images are from a z-series of 42 slices. Key: (A) slice 3; (B) slice 15; (C) slice 27; (D) slice 42. All bars represent 10  $\mu\text{m}$ .

antigen. As seen in Table 1, the aggregate size of the BODIPY FL labeled AC/AH complex in cell culture media was reduced by pretreatment of AH with phosphate. AP exhibited the smallest aggregate size in cell culture media.

Confocal microscopy revealed that AC adsorbed to PTAH-1, PTAH-2 and AP was also internalized by DCs. DCs internalized fragments of PTAH-1 and PTAH-2. However, the aggregates of AP were small enough (3  $\mu\text{m}$ ) that the DCs appeared to internalize whole aggregates of AP. In fact, as seen in Fig. 4, multiple 3  $\mu\text{m}$  aggregates of BODIPY FL labeled AC adsorbed to AP were seen within the DCs.

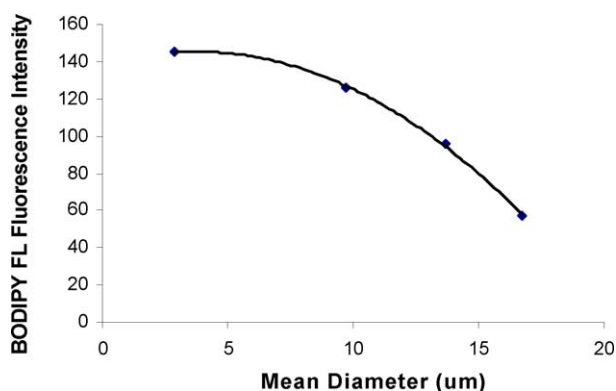


Fig. 5. Relationship between the degree of antigen internalization by dendritic cells and the mean diameter of the adjuvant aggregates in cell culture media.

When the DCs exposed to AC adsorbed to PTAH-1, PTAH-2 and AP were analyzed by flow cytometry, it was found that the fluorescence intensity within the DCs was inversely related to the aggregate size of the adjuvant in the cell culture media (Fig. 3). There does not appear to be any advantage in decreasing the aggregate size below the 3  $\mu\text{m}$  observed with AP since a plot of BODIPY FL fluorescence intensity versus the mean aggregate diameter becomes asymptotic at 3  $\mu\text{m}$  (Fig. 5).

#### 4. Summary

The results of the in vitro cell culture experiment indicate that internalization of antigen by DCs occurs rapidly by both macropinocytosis of eluted antigen and phagocytosis of adsorbed antigen. The internalization of adsorbed antigen was more efficient as the relative fluorescence intensity within the DCs was greater when adsorbed antigen was internalized in comparison to antigen in solution. In addition, the relationship between the size of the DCs and the size of the BODIPY FL labeled AC/adjuvant aggregates was important. Internalization was enhanced when the adjuvant aggregates were smaller than the DCs.

It is likely that both macropinocytosis and phagocytosis occur when the antigen elutes from the aluminum-containing adjuvant following administration. The relative contribution of macropinocytosis and phagocytosis to antigen internaliza-

tion by antigen presenting cells will likely depend on the rate of elution of antigen following administration. This model of antigen internalization is consistent with Chang et al.'s conclusion [4] that the degree of adsorption in interstitial fluid rather than the degree of adsorption in the vaccine correlates with antibody production.

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